LIBBY — PATIENT/CAREGIVER TRACK

osteosarcoma-mets-dll3-h7r2

Plain-language summary for patient and caregiver

Generated 2026-05-12

What this page is

You (or your clinician) asked Libby a narrow question. In metastatic osteosarcoma where there's a hint that DLL3 *and* PRAME might be present, what treatments would target either of those? You also said you'd consider high-risk options when the upside is meaningful, and that lens shaped how this report was put together.

One thing matters before anything else. Both signals you gave us are at the RNA level. RNA tells us each gene is being read inside the cell. It doesn't tell us the actual protein is sitting where the new drugs need to find it. For DLL3 the drugs need DLL3 protein on the outside of the cell. For PRAME the drugs need PRAME protein inside the cell *and* a specific immune marker called HLA-A*02:01 on the patient (because PRAME-targeting drugs are designed only to work when that marker is present). So the recommendations come in two parts. First, a set of confirmation tests that should happen up front. Then, options that only open up depending on which test result comes back positive.

The page is also scoped tightly. It only covers drugs that target DLL3 or PRAME, because those were the question. Plenty of other treatments exist for relapsed osteosarcoma, and those are a conversation for your oncologist, not this report.

What we know about your cancer

You're in your late teens or twenties with metastatic (stage IV) osteosarcoma, a cancer that started in bone. The plan assumes you've had standard frontline chemotherapy (the MAP regimen: methotrexate, doxorubicin, cisplatin), since that's the default for this disease, but your inputs didn't say how you responded or whether you've had anything since. Your day-to-day function looks like ECOG 1, which roughly means you're up and around but tire more easily than you used to. The two features driving this report are the DLL3 and PRAME RNA signals, with the caveat from the section above.

What you told us matters most

You said you accept high-risk, high-reward options when the possible upside is meaningful. You didn't name any toxicities you wanted to veto outright, and you flagged a clear preference for clinical trials over off-trial experimental use. Those signals are why first-in-disease trial options are even on the page.

The first step everyone agreed on

Before any treatment decision, get three tests done. They can run in parallel and the workup is the same one biopsy block plus one blood draw, so this is one trip, not three.

1. **DLL3 protein test on the tumor (IHC stain using the SP347 antibody).** Confirms whether DLL3 is actually on the surface of the cancer cells.
2. **PRAME protein test on the tumor (a different IHC stain).** Confirms whether the PRAME protein is being made.
3. **HLA-A*02:01 typing on a blood sample.** Confirms whether your immune system carries the marker that the PRAME-targeting drugs are built around.

In most cases your oncologist can order the IHC tests on stored tissue from a previous biopsy. If there isn't enough archived tissue, a fresh sample works too. None of these tests is toxic. Total turnaround is roughly one to three weeks for the IHC work; the HLA typing is faster. The cost is trivial compared to a single treatment cycle.

The workup decides which treatment options below are on the table at all. The two pathways are independent of each other, so you might have neither, one, or both.

If both DLL3 IHC *and* the PRAME / HLA pair come back negative, both pathways close and this page has no further treatment recommendations within scope. The conversation then shifts to standard relapsed-osteo options that your oncologist can take you through.

The options

There are four options, grouped by which test result has to be positive for them to be on the table. The DLL3 options come first because that was the original RNA signal in the case file; the PRAME options are equally serious if both PRAME tests are positive.

Option 1 — Tarlatamab via clinical trial NCT06788938 (DLL3 pathway)

Only available if the DLL3 IHC comes back positive. If the IHC is negative or below the trial's threshold, this option is off the table.

When the DLL3 test is positive, this is the lead option in the DLL3 pathway.

So what is it? Tarlatamab is a "bispecific T-cell engager." You get it as an infusion every two weeks. The drug works like a molecular clip. One end grips DLL3 on the cancer cell, the other end pulls in one of your own T cells to attack it. It's currently approved for small-cell lung cancer. NCT06788938 is a "basket" trial, meaning it accepts any DLL3-positive tumor (osteosarcoma included), provided the IHC confirms DLL3 is on the cell surface at high enough levels.

What might the upside look like? In small-cell lung cancer, where this drug is already approved, a recent randomized trial showed tarlatamab roughly cut the risk of death in half compared to chemotherapy. About 40 out of 100 patients had measurable tumor shrinkage. Whether any of that carries over to osteosarcoma is exactly the unanswered question this trial is designed to answer. I can't translate the lung-cancer numbers cleanly into an osteosarcoma expectation here, because no one has published that data yet and it would be misleading to pretend otherwise.

On the risk side, about half of patients get a reaction called "cytokine release syndrome." It's a flu-like immune flare; most cases are mild and roughly 1 in 100 are severe. About 1 in 10 patients have neurologic side effects. The first cycle requires inpatient monitoring, so you'd be in hospital while the step-up doses are given.

The open question on this option is cross-tumor translation. No osteosarcoma patient has been published as receiving tarlatamab yet. Whoever takes this trial is in the very first cohort. That's a real thing to weigh, not a footnote, and it's the trade against the strong lung-cancer track record.

Option 2 — SHR-4849 / IDE849 via clinical trial NCT07174583 (DLL3 pathway, considered with caveats)

This option carries caveats. Two locks have to clear before it's on the table: the DLL3 IHC has to come back positive AND the trial sponsor (IDEAYA) has to confirm osteosarcoma is on the basket eligibility list.

Why include it at all? It's a different *kind* of DLL3 drug from tarlatamab. Tarlatamab uses your T cells; SHR-4849 is an "antibody-drug conjugate," which means the antibody finds the DLL3 target and delivers a chemotherapy payload directly to the cancer cell. If the tarlatamab path is foreclosed (for example, the cytokine reaction is too rough to continue, or the cancer adapts and stops showing DLL3), an ADC keeps the DLL3 strategy in play through a different mechanism.

Why the caveats? Two reasons. First, no clinical results have been published for SHR-4849 yet. The phase 1/2 trial is ongoing, so the efficacy and toxicity profile will get sharper over time, but right now we're working from mechanism rather than data. Second, the basket eligibility list for this trial doesn't yet say "osteosarcoma" out loud. Pursuing it means your oncologist (or you) reaching IDEAYA medical affairs directly to confirm fit before going further. Both of those are real concerns to weigh against the second-mechanism hedge.

Option 3 — IMA203 (PRAME-TCR-T) via clinical trial NCT03686124 (PRAME pathway)

Only available if both PRAME IHC AND HLA-A*02:01 typing come back positive. Both have to be positive together. PRAME without HLA, or HLA without PRAME, doesn't unlock this option.

When both PRAME tests are positive, this is the lead option in the PRAME pathway.

What is it? IMA203 is a "TCR-T cell therapy." Your own T cells are collected, taken to a lab, given a custom receptor that recognizes the PRAME-derived peptide on HLA-A*02:01, multiplied, and infused back into you. NCT03686124 (the Immatics ACTengine basket) accepts any PRAME+/HLA-A*\02:01+ solid tumor, sarcoma cohorts included.

What might the upside look like? The published cohort data so far covers 28 patients across various solid tumors, including some sarcomas. About 54 out of 100 had measurable tumor shrinkage. That's a strong signal in heavily pretreated cancers, with responses lasting at least nine months in the patients who responded. Whether osteosarcoma specifically responds the same way is something the trial is set up to answer.

On the risk side, this is one of the more demanding options on the page. Almost everyone gets cytokine release syndrome (most cases mild, but about 1 in 10 are severe). About 1 in 4 have neurologic side effects. Before the cells are infused, you get a short course of two chemotherapy drugs (Cy/Flu) to make room for the new T cells; that drops your blood counts for several weeks. One published patient died from a complication of that pre-treatment chemotherapy. The manufacturing of your custom T cells takes four to six weeks, so this isn't a fast-start option.

The trade here is real. The published response rate is the strongest single number anywhere on this page, but the infrastructure burden, the toxicity profile, and the manufacturing wait are also the heaviest. The open question is whether the treating center has the apheresis capacity, lymphodepletion infrastructure, and CRS-monitoring protocols to run this safely.

Option 4 — IMC-P115C via clinical trial NCT07156136 (PRAME pathway, considered with caveats)

This option carries caveats. Three locks have to clear: PRAME IHC positive, HLA-A*02:01 typing positive, AND the trial sponsor (Immunocore) has to confirm osteosarcoma is on their basket eligibility list.

What is it? Like IMA203, IMC-P115C targets the PRAME peptide on HLA-A*02:01, but it does so with a smaller, off-the-shelf molecule called an "ImmTAC" rather than your own engineered T cells. You get it as a weekly infusion. The same broader drug class includes tebentafusp, which is an approved drug for a rare eye cancer, so the mechanism class itself is validated. IMC-P115C specifically is in phase 1 testing right now, so we don't yet have published efficacy numbers for it.

Why the caveats? Same shape as Option 2. No published clinical data for this specific drug yet, and the basket's osteosarcoma eligibility hasn't been spelled out, so pursuing it means contacting Immunocore medical affairs first. The reason to keep it on the page is the mechanism-class hedge: if the TCR-T path in Option 3 isn't reachable (insufficient infrastructure, manufacturing slot constraints, eligibility issue), an ImmTAC keeps the PRAME strategy in play through a different delivery.

What this report could not figure out

A few things the inputs didn't tell us, which would sharpen the plan:

- Were DLL3 and PRAME measured at the protein level (IHC) or only at RNA? The recommended first-step workup resolves this directly.
- What treatment have you actually had? The recommendations assume standard frontline MAP and nothing after that.
- What's your day-to-day function (performance status)? Most trials require ECOG 0 or 1.
- Where do you live, and how reachable are academic cancer centers? Each of these trials enrolls only at specific sites, and the PRAME-pathway ones in particular need a center that can do apheresis (Option 3) or weekly infusions with cytokine-release protocols.

Questions to ask your oncologist

1. Can we order all three tests at once: DLL3 IHC (SP347), PRAME IHC, and HLA-A*02:01 typing? They're the precondition for the rest of this conversation, and they can run on one biopsy block plus one blood draw.
2. Once the test results are in, can we walk through what each combination of results means for my options? "DLL3 positive only," "PRAME positive only with HLA," "both positive," and "neither" are four different conversations.
3. If DLL3 is positive, what's the realistic path to NCT06788938 from here? I'd want to understand the closest enrolling site, how long the wait usually is, and how the inpatient cycle-1 monitoring would actually look in practice.
4. If PRAME and HLA are both positive, what's the realistic path to NCT03686124? In particular, does the closest enrolling center have apheresis and the cytokine-release infrastructure? And how long is the manufacturing wait there?
5. If both pathways are foreclosed (negative on all three), what options do you usually consider for relapsed

osteosarcoma? This page doesn't cover those (it was scoped to DLL3 and PRAME), but they're real options and I'd want to understand them ahead of time.

6. What's your hospital's protocol for cytokine release syndrome and the neurologic reactions? Both Option 1 and Option 3 carry that risk, and how it's handled locally matters more than the published rates.
7. For the two trials that say "considered with caveats" (Options 2 and 4), would it be reasonable for your team to reach out to the sponsor's medical affairs to confirm whether osteosarcoma is in scope?
8. Have I already had any of the standard 2L+ osteosarcoma options? What was the response? Useful context for whoever's planning next steps.
9. Is there anything in my labs or imaging that would change which trials I'd be eligible for?

Sources

The recommendations cited the following published reports and clinical-trial records:

- Ahn et al. *N Engl J Med* 2023. Tarlatamab DeLLphi-301 in SCLC (PMID 37861218).
- Mountzios et al. *N Engl J Med* 2025. Tarlatamab DeLLphi-304 in SCLC (PMID 40454646).
- Wermke et al. *Lancet Oncol* 2024. IMA203 PRAME-TCR-T cross-tumor cohort (PMID 38821093).
- Hamid et al. *Cancer Discov* 2024. Brenetafusp PRAME ImmTAC class precedent (PMID 39007852).
- Iura et al. 2017. PRAME expression frequency in osteosarcoma (PMID 28315425).
- Yao et al. *Oncologist* 2022. DLL3 as a therapeutic target review (PMID 35983951).
- Wang et al. *J Exp Clin Cancer Res* 2018. Notch3/DLL3 axis in osteosarcoma stemness (PMID 29475441).
- Trial registry NCT06788938 (UCCC-01 / UCLA L-10 tarlatamab basket).
- Trial registry NCT07174583 (IDEAYA SHR-4849 / IDE849 pan-tumor DLL3 basket).
- Trial registry NCT03686124 (Immatics ACTengine pan-solid PRAME TCR-T basket).
- Trial registry NCT07156136 (Immunocore IMC-P115C pan-tumor PRAME ImmTAC).